

VALLEYVIEW AGDM, AB, Canada

WMO: 712770

Lat: 55.0981N

Lon: 117.1992W

Elev: 698

StdP: 93.22

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.8	-27.6	-34.4	0.2	-30.7	-31.0	0.2	-27.3	11.8	1.6	10.5	0.5	2.9	170	0.693

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	27.7	16.3	25.9	15.6	24.2	14.9	18.0	25.1	16.9	23.7	15.9	22.2	3.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.8	20.5	14.3	11.1	19.2	13.3	10.4	18.0	53.5	24.9	50.0	23.6	47.0	22.3	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.5	7.6	-35.5	31.2	4.1	2.0	-38.4	32.6	-40.8	33.8	-43.1	34.9	-46.1	36.4
			-35.6	19.7	4.1	1.3	-38.5	20.7	-40.9	21.4	-43.2	22.1	-46.1	23.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-9.7	-8.8	-3.6	3.6	10.3	14.2	16.6	15.2	10.6	4.3	-4.9	-10.5
	DBStd	11.51	8.87	8.30	7.30	5.05	4.40	3.05	2.81	3.34	4.47	4.94	7.04	8.07
	HDD10.0	3122	609	528	423	196	49	3	0	2	45	185	447	634
	HDD18.3	5565	868	761	681	441	249	129	69	107	234	437	697	893
	CDD10.0	630	0	0	0	5	59	129	204	163	62	7	0	0
	CDD18.3	29	0	0	0	0	1	4	14	9	2	0	0	0
	CDH23.3	562	0	0	0	3	38	87	231	151	50	2	0	0
	CDH26.7	109	0	0	0	0	6	12	52	31	9	0	0	0
(14) Wind	WSAvg	3.9	4.2	3.8	3.9	4.1	4.0	3.7	3.4	3.3	3.8	4.0	4.0	4.2
(15) Precipitation	PrecAvg	486	32	19	19	20	49	85	78	60	40	26	29	26
	PrecMax	688	93	41	36	43	98	151	122	132	118	54	73	50
	PrecMin	320	8	5	5	3	18	43	27	22	4	7	8	5
	PrecStd	91	23	9	10	10	24	29	30	28	26	12	19	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	7.8	12.1	21.6	27.2	28.3	31.0	29.8	28.0	21.5	12.3	6.6
		MCWB	3.2	3.2	5.6	10.3	13.4	16.2	17.9	17.1	15.6	12.0	6.0	3.0
	2%	DB	5.6	5.5	9.2	17.2	23.9	25.8	27.9	26.9	24.5	16.5	8.0	4.3
		MCWB	2.4	1.8	3.8	7.9	12.3	14.9	17.1	16.6	14.3	9.5	4.0	1.6
	5%	DB	3.9	3.7	7.2	13.9	21.5	23.8	25.8	24.8	21.4	13.5	5.3	2.3
		MCWB	1.4	0.7	2.7	6.4	10.9	14.0	16.4	15.4	13.0	7.5	2.4	-0.1
	10%	DB	2.0	1.7	5.2	11.5	19.2	21.7	23.9	22.8	18.3	11.3	3.4	0.4
		MCWB	0.2	-0.7	1.5	5.1	9.9	13.1	15.7	14.9	11.4	6.2	1.0	-1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.5	6.1	10.7	14.5	17.5	19.9	18.9	16.6	12.3	6.5	3.3
		MCDB	6.8	6.8	11.5	20.7	24.4	25.1	26.8	26.4	25.5	20.1	11.1	6.0
	2%	WB	2.6	2.0	4.1	8.6	12.8	16.1	18.5	17.7	15.0	9.9	4.3	1.7
		MCDB	5.1	5.1	8.4	16.0	21.7	23.4	25.4	24.8	22.9	15.4	7.4	4.0
	5%	WB	1.6	0.9	2.9	6.7	11.7	15.0	17.2	16.6	13.6	8.2	2.7	0.1
		MCDB	3.8	3.3	6.9	13.1	19.7	21.7	23.9	23.0	20.1	12.5	5.0	2.0
	10%	WB	0.0	-0.5	1.7	5.4	10.6	14.0	16.3	15.5	12.2	6.7	1.2	-1.5
		MCDB	2.0	1.7	5.0	10.8	17.9	20.3	22.4	21.2	17.4	10.5	3.1	0.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.9	10.7	10.9	12.7	11.8	12.3	12.4	11.9	9.7	7.7	7.8
		MCDBR	8.9	10.9	11.9	15.2	16.9	15.8	16.0	16.0	17.0	13.5	9.2	9.5
		MCWBR	6.7	7.8	7.5	8.2	7.5	6.8	6.8	6.8	7.6	6.3	6.3	7.5
	5% WB	MCDBR	8.4	10.4	11.4	14.4	15.1	13.5	14.2	14.5	15.8	12.9	8.8	9.6
		MCWBR	6.5	7.6	7.3	8.1	7.1	6.4	6.8	6.8	7.3	7.6	6.4	7.8
(40) Clear-Sky Solar Irradiance	taub		0.241	0.249	0.286	0.330	0.342	0.348	0.349	0.348	0.310	0.283	0.251	0.236
	taud		2.364	2.433	2.417	2.390	2.409	2.410	2.436	2.422	2.540	2.584	2.461	2.283
	Ebn at Noon		735	868	900	896	902	897	889	867	865	818	722	641
	Edh at Noon		49	66	86	103	107	109	104	99	77	58	45	42
(44) All-Sky Solar Radiation	RadAvg		0.79	1.78	3.27	4.63	5.49	5.90	5.83	4.82	3.30	1.81	0.84	0.57
	RadStd		0.07	0.14	0.27	0.31	0.41	0.47	0.40	0.38	0.29	0.20	0.07	0.05

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page